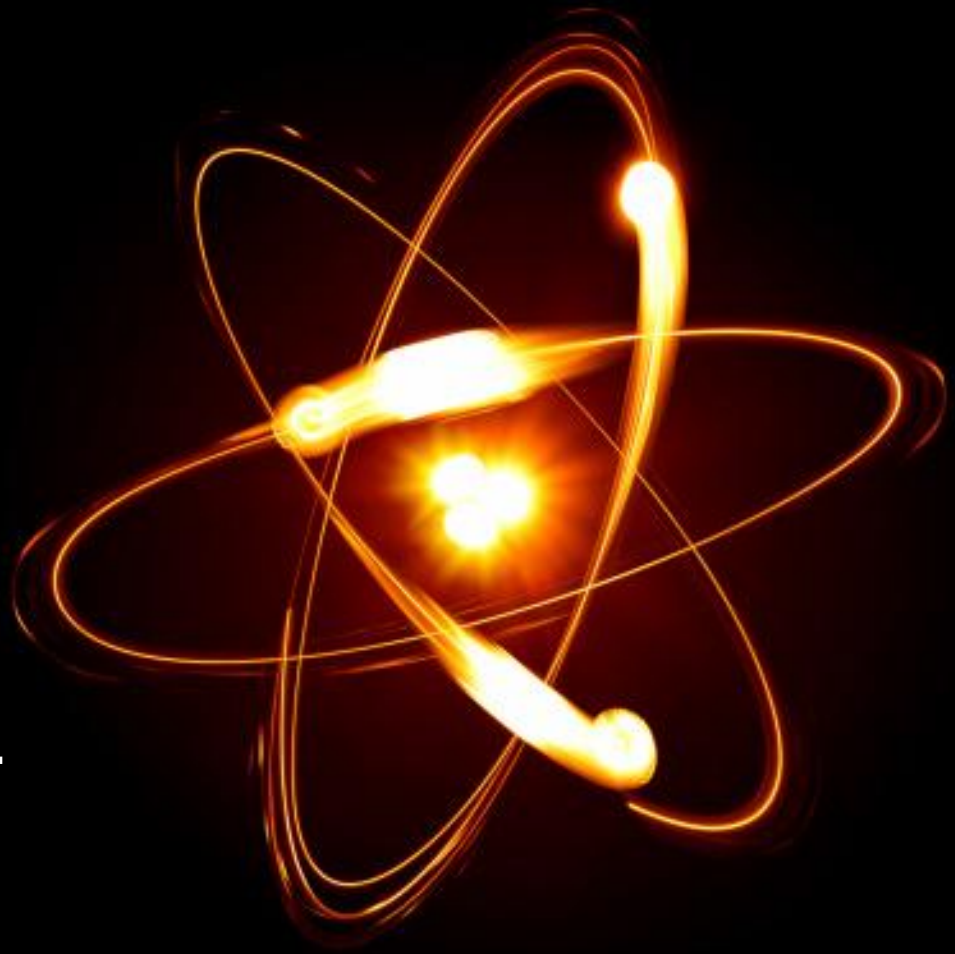
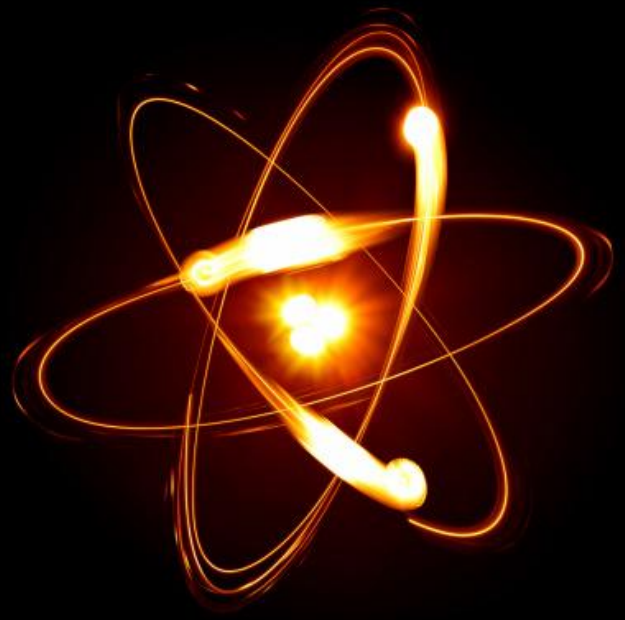
Chapter 22.

Addition of angular momenta. General method



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Quantum Physics



Chapter 22.

Addition of general momenta. General method

- Review of angular momenta
- Statement of the problem
- Addition of angular momenta
- Eigenvalues of J^2 and J_z
- Eigenstates of J^2 and J_z
- Addition of two angular momenta with s= ½.
- Clebsh-Gordan coefficients

Review of angular momenta

System with state space $\mathcal{E}$, and angular momentum $\vec{J} = (J_x, J_y, J_z)$

We know that we can construct a standard basis $\{|k,j,m\rangle\}$ made of common eigenvectors of J^2 and J_z

$$J^2|k,j,m\rangle = j(j+1)\hbar^2|k,j,m\rangle \qquad J_z|k,j,m\rangle = m\hbar|k,j,m\rangle \qquad J_\pm|k,j,m\rangle = \hbar\sqrt{j(j+1)-m(m\pm1)}\,|k,j,m\pm1\rangle$$

$\mathcal{E}(k,j)$ is the vector space generated by $|k,j,m\rangle$ with fixed k and j.

- Its dimension is $(2j+1)$
- $\mathcal{E}(k,j)$ is invariant under the action of J^2, J_z, $J_\pm$ or any function $F(\vec{J})$.
- Inside each $\mathcal{E}(k,j)$, the matrix elements of any function $F(\vec{J})$ are independent of k.

The total state space, $\mathcal{E}$ can be written as a direct sum of the previous subspaces, $\mathcal{E} = \sum_{\otimes} \mathcal{E}(k,j)$

Statement of the problem

Physical system formed by the union of two subsystems: (1) and (2)

Subsystem (1), with state space $\mathcal{E}_1$, standard basis $\{|k_1, j_1, m_1\rangle\}$ made of common eigenvectors of J_1^2 and J_{1z}

$$J_1^2|k_1, j_1, m_1\rangle = j_1(j_1+1)\hbar^2|k_1, j_1, m_1\rangle \qquad J_{1z}|k_1, j_1, m_1\rangle = m_1\hbar|k_1, j_1, m_1\rangle$$

$$J_{1\pm}|k_1, j_1, m_1\rangle = \hbar\sqrt{j_1(j_1+1) - m_1(m_1 \pm 1)}|k_1, j_1, m_1 \pm 1\rangle$$

Subsystem (2), with state space $\mathcal{E}_2$, standard basis $\{|k_2, j_2, m_2\rangle\}$ made of common eigenvectors of J_2^2 and J_{2z}

$$J_2^2|k_2, j_2, m_2\rangle = j_2(j_2+1)\hbar^2|k_2, j_2, m_2\rangle \qquad J_{2z}|k_2, j_2, m_2\rangle = m_2\hbar|k_2, j_2, m_2\rangle$$

$$J_{2\pm}|k_2, j_2, m_2\rangle = \hbar\sqrt{j_2(j_2+1) - m_2(m_2 \pm 1)}|k_2, j_2, m_2 \pm 1\rangle$$

State space of the global system $\mathcal{E} = \mathcal{E}_1 \otimes \mathcal{E}_2$

Statement of the problem

Basis of the global system: Tensor product of basis taken in $\mathcal{E}_1$ and $\mathcal{E}_2$

$$|k_1, k_2;\ j_1, j_2;\ m_1, m_2\rangle = \{|k_1, j_1, m_1\rangle\} \otimes \{|k_2, j_2, m_2\rangle\}$$

Remember that

$$\mathcal{E}_1 = \sum_{\otimes} \mathcal{E}_1(k_1, j_1) \qquad \mathcal{E}_2 = \sum_{\otimes} \mathcal{E}_2(k_2, j_2)$$

Then $\mathcal{E} = \sum_{\otimes} \mathcal{E}(k_1, k_2;\ j_1, j_2)$ where

$$\mathcal{E}(k_1, k_2;\ j_1, j_2) = \mathcal{E}_1(k_1, j_1) \otimes \mathcal{E}_2(k_2, j_2)$$

is the tensor product of a space $\mathcal{E}_1(k_1, j_1)$ and a space $\mathcal{E}_2(k_2, j_2)$

Statement of the problem

The total angular momentum of the global system is $\vec{J} = \vec{J}_1 + \vec{J}_2$ and we know that

- $J_{1x}, J_{1y}, J_{1z}, J_{2x}, J_{2y}, J_{2z}$ commute with J_1^2 and J_2^2
- J_x, J_y, J_z commute with J^2. It is an angular momentum

Moreover

- $[J_z, J_1^2] = 0$ $\quad$ $[J_z, J_2^2] = 0$
- $[J^2, J_1^2] = 0$ $\quad$ $[J^2, J_2^2] = 0$
- $[J_z, J_{1z}] = 0$ $\quad$ $[J_z, J_{2z}] = 0$
- But J^2 does NOT commute with J_{1z} and J_{2z}
- $J^2 = J_1^2 + J_2^2 + 2J_{1z}J_{2z} + J_{1+}J_{2-} + J_{1-}J_{2+}$

Statement of the problem. Change of basis

A vector $|k_1, k_2;\ j_1, j_2;\ m_1, m_2\rangle$ is a simultaneous eigenvector of J_1^2, J_2^2, J_{1z} and J_{2z}

But we know that J_1^2, J_2^2, J^2 and J_z commute between them. So **we construct an orthonormal basis of $\mathcal{E}$ made by common eigenvectors of these observables**

Remember that $\mathcal{E} = \sum_{\otimes} \mathcal{E}(k_1, k_2;\ j_1, j_2)$ and each subspace $\mathcal{E}(k_1, k_2;\ j_1, j_2)$ is invariant under the action of any operator that is a function of $\vec{J}_1$ or $\vec{J}_2$.

Therefore the observables J^2 and J_z , **which we want to diagonalize,** have non-zero matrix elements only between vectors belonging to the same subspace $\mathcal{E}(k_1, k_2;\ j_1, j_2)$. Moreover, those matrix elements are independent of k_1 and k_2

In fact, the problem of diagonalizing J^2 and J_z is the same inside all the subspaces $\mathcal{E}(k_1, k_2;\ j_1, j_2)$ which have the same values of j_1 and j_2.

Statement of the problem

The problem reduces to a change of basis inside each of the subspaces $\mathcal{E}(k_1, k_2;\, j_1, j_2)$, which are of dimension $(2j_1 + 1)(2j_2 + 1)$

The diagonalization process inside each subspace $\mathcal{E}(k_1, k_2;\, j_1, j_2)$, (with dimension $(2j_1 + 1)(2j_2 + 1)$) yields the eigenvectors (and eigenvalues) common to J_1^2, J_2^2, J^2 and J_z

We denote the eigenvectors as $|j_1, j_2, J, M\rangle \equiv |J, M\rangle$ where $j_1(j_1 + 1)\hbar^2$, $j_2(j_2 + 1)\hbar^2$, $J(J + 1)\hbar^2$ and $M\hbar$ are the respective eigenvalues.

So, in the end, the problem boils down to:

- Given j_1 and j_2, what are the possible values of J?
- How can the eigenvectors of J^2 and J_z , belonging to $\mathcal{E}(k_1, k_2;\, j_1, j_2)$, be expanded in terms of the $|k_1, k_2;\, j_1, j_2\rangle$?

Addition of two angular momenta[6]

Adding two angular momenta $\vec{J}_1$ and $\vec{J}_2$ $\left(\vec{J} = \vec{J}_1 + \vec{J}_2\right)$ amounts basically to finding a basis of the state space made up by common eigenvectors of J_1^2, J_2^2, J^2 and $J_z = J_{1z} + J_{2z}$, namely the vectors $|j_1, j_2, J, M\rangle \equiv |J, M\rangle$

These are linear combinations of the eigenvectors of J_1^2, J_2^2, J_{1z} and J_{2z}, namely, $|\, j_1, j_2, m_1, m_2\rangle$, which are the vectors of the old basis.

$$|J, M\rangle = \sum \underbrace{\alpha(j_1, j_2, m_1, m_2)}_{*} |\, j_1, j_2, m_1, m_2\rangle$$

where the coefficients * are known as Clebsch-Gordan coefficients[5]

[5]Complement B Chap X Cohen
[6]Complement C, Chap X, Cohen

Eigenvalues of J^2 and J_z

Finding these eigenvectors also implies finding the eigenvalues, that is, finding the possible values of *J* and *M*.

Rules

The possible values of *J*, when adding up two angular momenta with associated values j_1 and j_2, are

$$J = j_1 + j_2, j_1 + j_2 - 1, j_1 + j_2 - 2, \dots, |j_1 - j_2|$$

The possible values of *M* are obtained from the fact that $\vec{J}$ is an angular momentum, so

$$M = -J, -J + 1, \dots, J - 1, J$$

Where do these rules come from?

Eigenvalues of J_z

The states $|j_1, j_2, m_1, m_2\rangle$ are eigenstates of J_z with eigenvalue $m_1 + m_2$:

$$J_z|j_1, j_2, m_1, m_2\rangle = (J_{1z} + J_{2z})(|j_1, m_1\rangle \otimes |j_2, m_2\rangle) = (J_{1z}|j_1, m_1\rangle) \otimes |j_2, m_2\rangle + |j_1, m_1\rangle \otimes (J_{2z}|j_2, m_2\rangle) =$$

$$= (m_1 + m_2)\hbar|j_1, j_2, m_1, m_2\rangle = M\hbar|j_1, j_2, m_1, m_2\rangle$$

Since $-j_1 \leq m_1 \leq j_1$ and $-j_2 \leq m_2 \leq j_2$, we deduce that the maximum and minimum possible values of M are $j_1 + j_2$ and $-j_1 - j_2$ respectively. ⇒ Consequently the maximum possible value of J is $j_1 + j_2$.

The state with $m_1 = j_1, m_2 = j_2$ is the ONLY eigenstate of J_z with eigenvalue $M = (j_1 + j_2)\hbar$. Therefore it must be an eigenvector of J^2 with eigenvalue $J(J+1)\hbar^2$, where $J = j_1 + j_2$.

Iterated applications of J_- on this state lead to eigenstates of J^2 and J_z with the same $J = j_1 + j_2$ and with $M = j_1 + j_2 - 1, j_1 + j_2 - 2, \ldots, -j_1 - j_2 + 1, -j_1 - j_2$

Eigenvalues of J_z

Degeneracy of the M-values (case $j_1 = 2$, $j_2 = 1$)

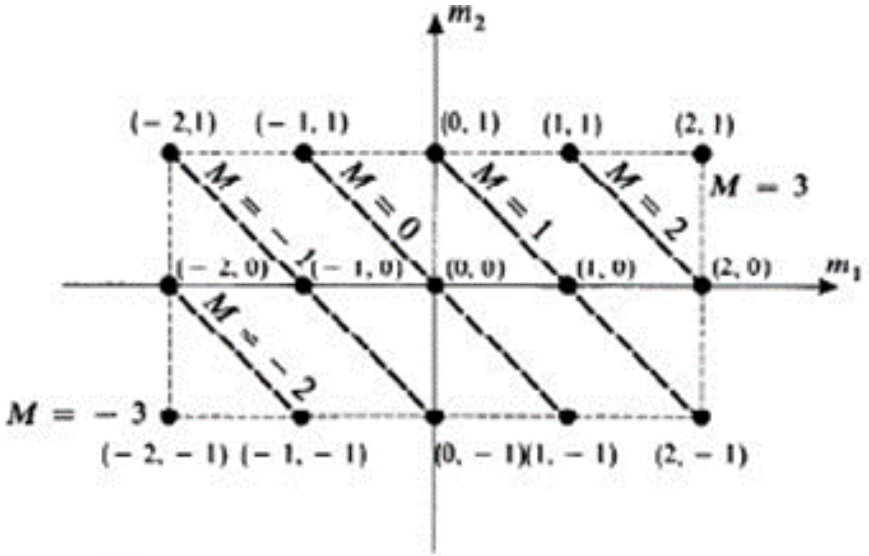


FIGURE 1

Pairs of possible values (m_1, m_2) for the kets $|\, j_1, j_2; m_1, m_2 \rangle$. We have chosen the case in which $j_1 = 2$ and $j_2 = 1$. The points associated with a given value of $M = m_1 + m_2$ are situated on a straight line of slope -1 (dashed lines).

With each vector $|j_1 = 2, j_2 = 1, m_1, m_2\rangle$ we associate a point whose abscissa is m_1 and ordinate is m_2. The figure represents the fifteen points associated with the basis vectors $|j_1 = 2, j_2 = 1, m_1, m_2\rangle$.

All points on the same dashed line correspond to the same value $M = m_1 + m_2$. The number of such points gives the degeneracy of that M-value.

Eigenstates of J^2 and J_z

There are two eigenstates with $M = j_1 + j_2 - 1$. One of them corresponds to $J = j_1 + j_2$ just obtained. An eigenstate with the same M orthogonal to this one, must then correspond to $J = j_1 + j_2 - 1$, since it is the only one left.

Iterated application of J_- will lead to all the eigenstates of J^2 with $J = j_1 + j_2 - 1$.

And so on ...

When we arrive to the state with $M = J = |j_1 - j_2|$ and after iterated application of J_- to all the states with $J = |j_1 - j_2|$ we exhaust all the states and then we are done.

With each J-value, there is an associated subspace $\mathcal{E}(J)$

Each subspace $\mathcal{E}(J)$ is invariant under the action of J^2, J_z, J_- and J_+.

The $\mathcal{E}(j_1, j_2)$ is a direct sum of subspaces $\mathcal{E}(J)$, i.e. $\mathcal{E}(j_1, j_2) = \sum_{\otimes} \mathcal{E}(J)$

Addition of two angular momenta with s=1/2

α. *The subspace $\mathcal{E}(S = 1)$*

The ket $|+, +\rangle$ is, in the state space $\mathcal{E} = \mathcal{E}(1/2, 1/2)$, the only eigenvector of S_z associated with $M = 1$. Since $\mathbf{S}^2$ and S_z commute, and the value $M = 1$ is not degenerate, $|+, +\rangle$ must also be an eigenvector of $\mathbf{S}^2$ (§ D-3-a of chapter II). According to the reasoning of §C-3-a, the corresponding value of S must be 1. Therefore, we can choose the phase of the vector $|S = 1, M = 1\rangle$ such that:

$$|1, 1\rangle = |+, +\rangle \tag{C-43}$$

It is then easy to find the other states of the triplet, since we know from the general theory of angular momentum that:

$$\begin{aligned} S_- |1, 1\rangle &= \hbar\sqrt{1(1+1) - 1(1-1)}\,|1, 0\rangle \\ &= \hbar\sqrt{2}\,|1, 0\rangle \end{aligned} \tag{C-44}$$

Consequently:

$$|1, 0\rangle = \frac{1}{\hbar\sqrt{2}} S_- |+, +\rangle \tag{C-45}$$

To calculate $|1, 0\rangle$ explicitly in the $\{|\varepsilon_1, \varepsilon_2\rangle\}$ basis, it suffices to recall that definition (B-3) of the total spin $\mathbf{S}$ implies:

$$S_- = S_{1-} + S_{2-} \tag{C-46}$$

Addition of two angular momenta with s=1/2

We then obtain:

$$
\begin{aligned}
|1,0\rangle &= \frac{1}{\hbar\sqrt{2}}(S_{1-}+S_{2-})|+,+\rangle \\
&= \frac{1}{\hbar\sqrt{2}}[\hbar|-,+\rangle+\hbar|+,-\rangle] \\
&= \frac{1}{\sqrt{2}}[|-,+\rangle+|+,-\rangle]
\end{aligned}
$$

Finally, we can again apply S_- to $|1,0\rangle$, that is, $(S_{1-}+S_{2-})$ to expression (C-47). This yields:

$$
\begin{aligned}
|1,-1\rangle &= \frac{1}{\hbar\sqrt{2}}S_-|1,0\rangle \\
&= \frac{1}{\hbar\sqrt{2}}(S_{1-}+S_{2-})\frac{1}{\sqrt{2}}[|-,+\rangle+|+,-\rangle] \\
&= \frac{1}{2\hbar}[\hbar|-,-\rangle+\hbar|-,-\rangle] \\
&= |-,-\rangle
\end{aligned}
$$

Of course, this last result could have been obtained directly, using an argument analogous to the one applied above to $|+,+\rangle$. However, the preceding calculation has a slight advantage : it enables us, in accordance with the general conventions

Addition of two angular momenta with s=1/2

β. *The state* $|\, S = 0, M = 0 \,\rangle$

The only vector $|\, S = 0, M = 0 \,\rangle$ of the subspace $\mathscr{E}(S = 0)$ is determined, to within a constant factor, by the condition that it must be orthogonal to the three vectors $|\, 1, M \,\rangle$ which we have just constructed.

Since it is orthogonal to $|\, 1, 1 \,\rangle = |\, +, + \,\rangle$ and $|\, 1, -1 \,\rangle = |\, -, - \,\rangle$, $|\, 0, 0 \,\rangle$ must be a linear combination of $|\, +, - \,\rangle$ and $|\, -, + \,\rangle$:

$$|\, 0, 0 \,\rangle = \alpha\, |\, +, - \,\rangle + \beta\, |\, -, + \,\rangle$$

which will be normalized if:

$$\langle 0, 0 \,|\, 0, 0 \rangle = |\alpha|^2 + |\beta|^2 = 1$$

We now insist that its scalar product with $|\, 1, 0 \,\rangle$ [*cf.* (C-47)] be zero:

$$\frac{1}{\sqrt{2}}(\alpha + \beta) = 0$$

Addition of two angular momenta with s=1/2

The coefficients α and β are therefore equal in absolute value and of opposite sign. With (C-50) taken into account, this fixes them to within a phase factor:

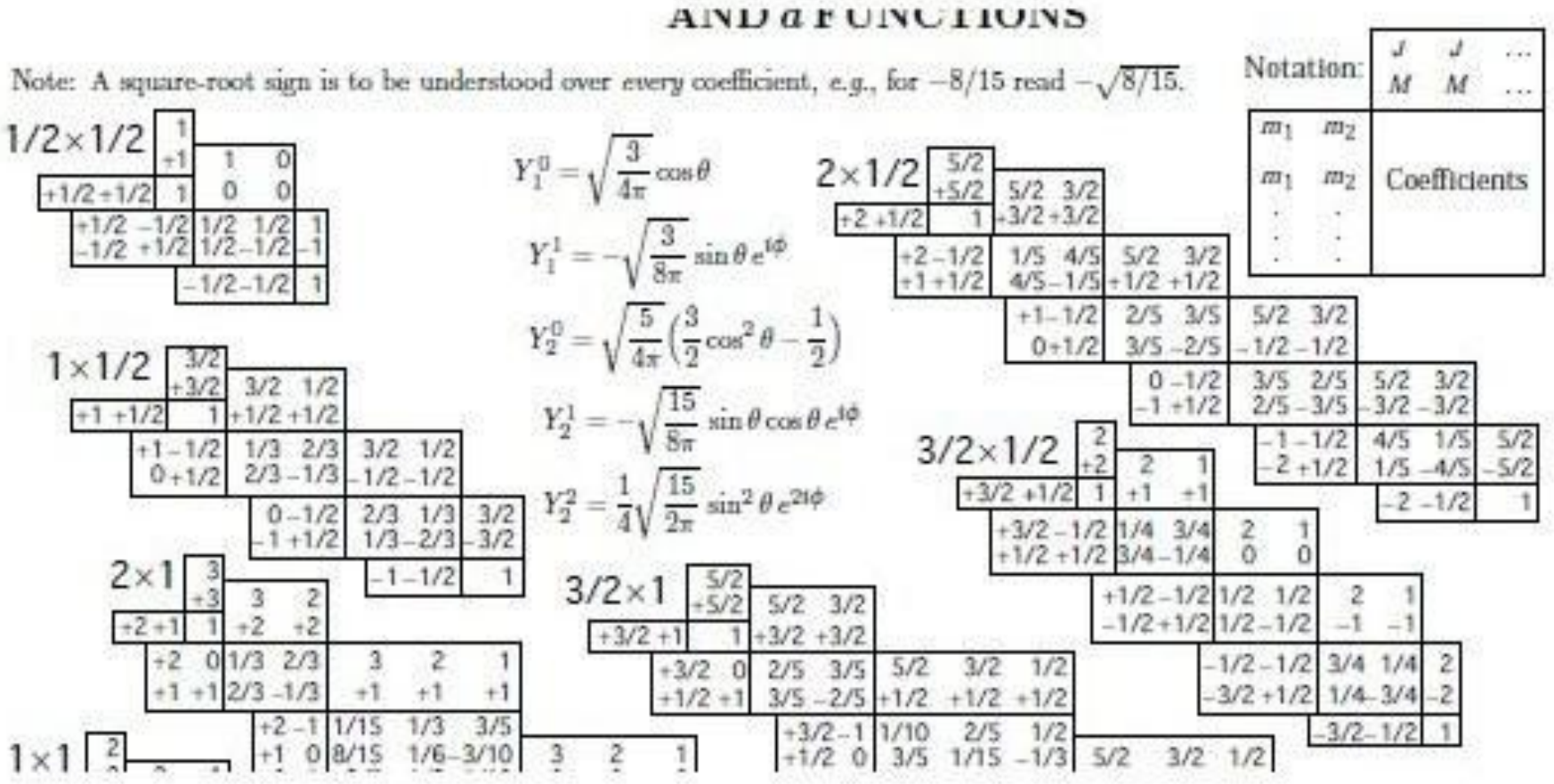
$$\alpha = -\beta = \frac{1}{\sqrt{2}} e^{i\chi}$$

where χ is any real number. We shall choose $\chi = 0$, which yields:

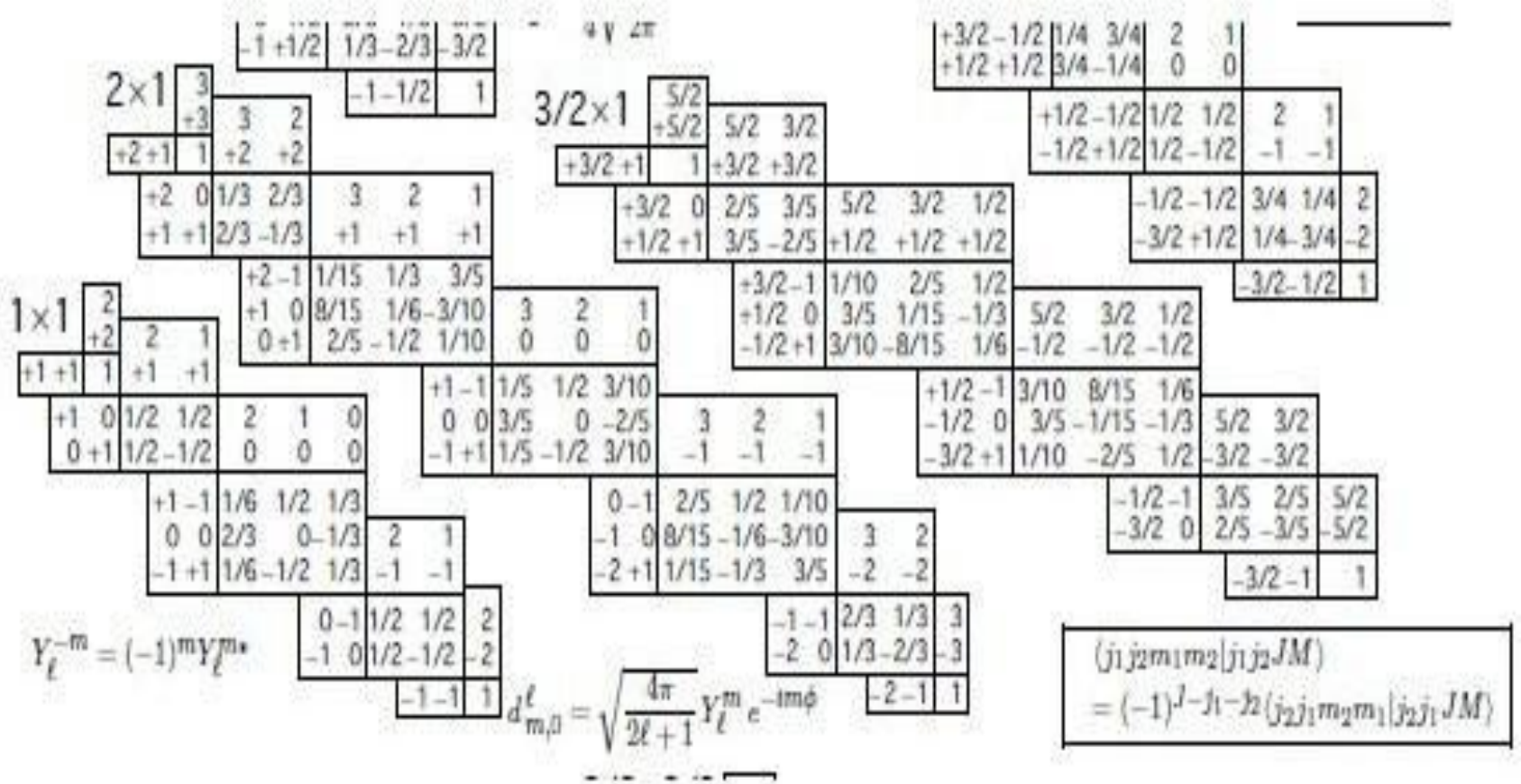
$$|0, 0\rangle = \frac{1}{\sqrt{2}} [|+, -\rangle - |-, +\rangle]$$

Thus we have calculated the four vectors $|S, M\rangle$ without explicitly having had to write the matrix which represents $\mathbf{S}^2$ in the $\{|\varepsilon_1, \varepsilon_2\rangle\}$ basis.

Clebsch-Gordan Coefficients

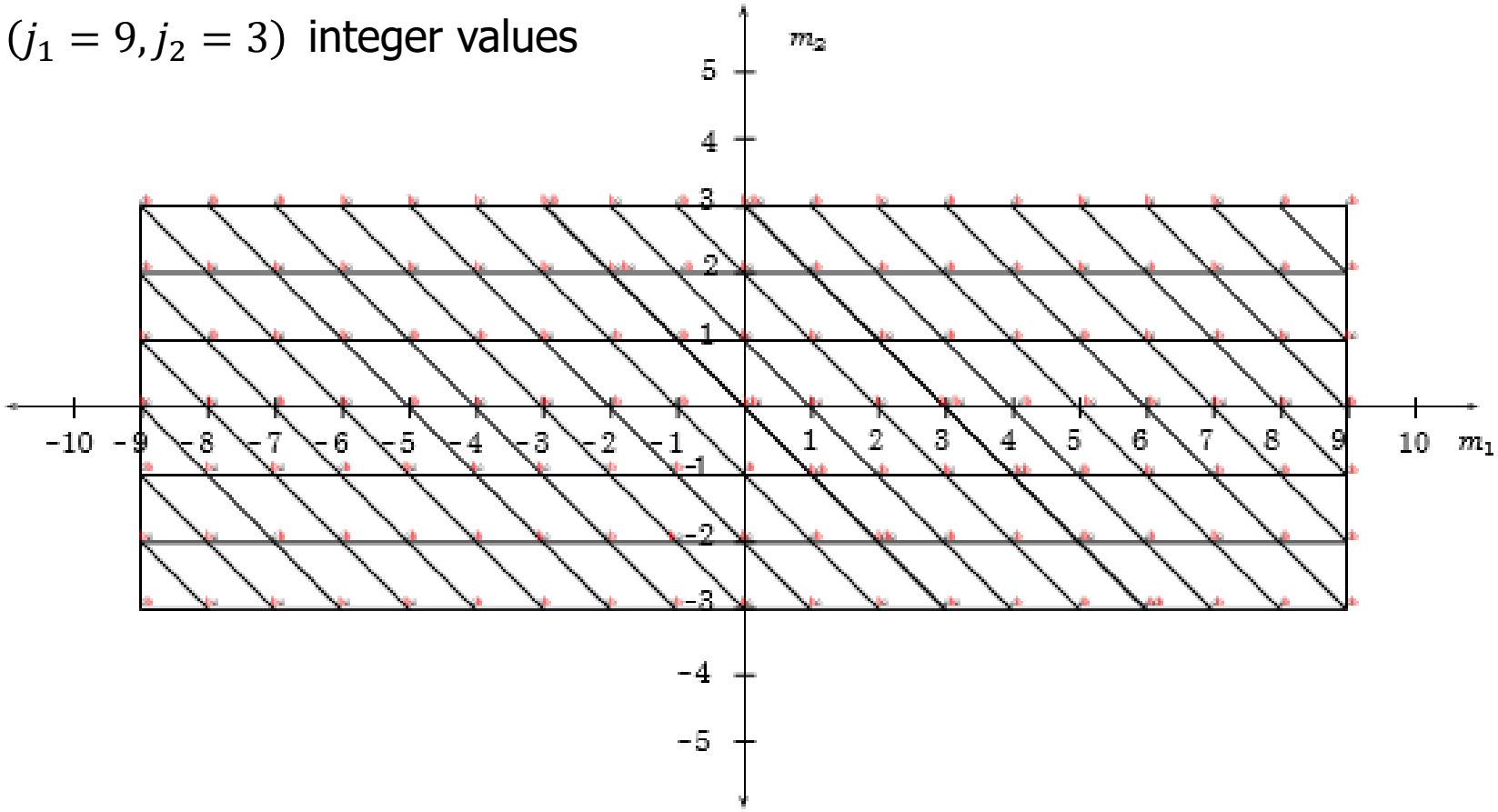


Clebsch-Gordan Coefficients



Addition of two angular momenta

$j_1 > j_2$ $(j_1 = 9, j_2 = 3)$ integer values



Addition of two angular momenta

$j_1 > j_2$ $(j_1 = {}^{17}/_2, j_2 = {}^{7}/_2)$ half-integer values

